

7.5 Practice - Complex Fractions

Solve.

$$1) \frac{1 + \frac{1}{x}}{1 - \frac{1}{x^2}}$$

$$2) \frac{\frac{1}{y^2} - 1}{1 + \frac{1}{y}}$$

$$3) \frac{a - 2}{\frac{4}{a} - a}$$

$$4) \frac{\frac{25}{a} - a}{5 + a}$$

$$5) \frac{\frac{1}{a^2} - \frac{1}{a}}{\frac{1}{a^2} + \frac{1}{a}}$$

$$6) \frac{\frac{1}{b} + \frac{1}{2}}{\frac{4}{b^2 - 1}}$$

$$7) \frac{2 - \frac{4}{x+2}}{5 - \frac{10}{x+2}}$$

$$8) \frac{4 + \frac{12}{2x-3}}{5 + \frac{15}{2x-3}}$$

$$9) \frac{\frac{3}{2a-3} + 2}{\frac{-6}{2a-3} - 4}$$

$$10) \frac{\frac{-5}{b-5} - 3}{\frac{10}{b-5} + 6}$$

$$11) \frac{\frac{x}{x+1} - \frac{1}{x}}{\frac{x}{x+1} + \frac{1}{x}}$$

$$12) \frac{\frac{2a}{a-1} - \frac{3}{a}}{\frac{-6}{a-1} - 4}$$

$$13) \frac{\frac{3}{x}}{\frac{9}{x^2}}$$

$$14) \frac{\frac{x}{3x-2}}{\frac{x}{9x^2-4}}$$

$$15) \frac{\frac{a^2-b^2}{4a^2b}}{\frac{a+b}{16ab^2}}$$

$$16) \frac{1 - \frac{1}{x} - \frac{6}{x^2}}{1 - \frac{4}{x} + \frac{3}{x^2}}$$

$$17) \frac{1 - \frac{3}{x} - \frac{10}{x^2}}{1 + \frac{11}{x} + \frac{18}{x^2}}$$

$$18) \frac{\frac{15}{x^2} - \frac{2}{x} - 1}{\frac{4}{x^2} - \frac{5}{x} + 4}$$

$$19) \frac{1 - \frac{2x}{3x-4}}{x - \frac{32}{3x-4}}$$

$$20) \frac{1 - \frac{12}{3x+10}}{x - \frac{8}{3x+10}}$$

$$21) \frac{x - 1 + \frac{2}{x-4}}{x + 3 + \frac{6}{x-4}}$$

$$22) \frac{x - 5 - \frac{18}{x+2}}{x + 7 + \frac{6}{x+2}}$$

$$23) \frac{x-4+\frac{9}{2x+3}}{x+3-\frac{5}{2x+3}}$$

$$24) \frac{\frac{1}{a}-\frac{3}{a-2}}{\frac{2}{a}+\frac{5}{a-2}}$$

$$25) \frac{\frac{2}{b}-\frac{5}{b+3}}{\frac{3}{b}+\frac{3}{b+3}}$$

$$26) \frac{\frac{1}{y^2}-\frac{1}{xy}-\frac{2}{x^2}}{\frac{1}{y^2}-\frac{3}{xy}+\frac{2}{x^2}}$$

$$27) \frac{\frac{2}{b^2}-\frac{5}{ab}-\frac{3}{a^2}}{\frac{2}{b^2}+\frac{7}{ab}+\frac{3}{a^2}}$$

$$28) \frac{\frac{x-1}{x+1}-\frac{x+1}{x-1}}{\frac{x-1}{x+1}+\frac{x+1}{x-1}}$$

$$29) \frac{\frac{y}{y+2}-\frac{y}{y-2}}{\frac{y}{y+2}+\frac{y}{y-2}}$$

$$30) \frac{\frac{x+1}{x-1}-\frac{1-x}{1+x}}{\frac{1}{(x+1)^2}+\frac{1}{(x-1)^2}}$$

Simplify each of the following fractional expressions.

$$31) \frac{x^{-2}-y^{-2}}{x^{-1}+y^{-1}}$$

$$32) \frac{x^{-2}y+xy^{-2}}{x^{-2}-y^{-2}}$$

$$33) \frac{x^{-3}y-xy^{-3}}{x^{-2}-y^{-2}}$$

$$34) \frac{4-4x^{-1}+x^{-2}}{4-x^{-2}}$$

$$35) \frac{x^{-2}-6x^{-1}+9}{x^2-9}$$

$$36) \frac{x^{-3}+y^{-3}}{x^{-2}-x^{-1}y^{-1}+y^{-2}}$$